

ASU Macro Theory Qualifier

Module 2: OLG Model

PhD Qual Lecture Notes Builder Packet

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Orientation

This packet is designed for first-round ASU macro qualifying-exam preparation. It is not a textbook chapter. The aim is to make the OLG workflow automatic under time pressure:

household problem $\rightarrow$ FOCs/log rule $\rightarrow$ aggregate savings $\rightarrow$ capital law $\rightarrow$ steady state or BGP $\rightarrow$ comparative s

The target weaknesses are formulation and setup, economic intuition, steady state versus balanced-growth-path analysis, transitional dynamics, and comparative statics.

How to use this packet

Use Sections 7–8 as the derivation core. Use Section 11 to recognize past-qual mutations. Use Sections 13–14 for timed practice. A good study plan is two 3-hour blocks: one block for canonical derivations and one block for past-qual variants.

1 Source Inventory

1.1 Governing project documents

Source	Type	Topics and use	Extraction notes
/mnt/data/SKILL.md	Skill spec	Required workflow: inventory, pattern analysis, blueprint, full packet, exercises, answer sketches, quality audit.	Read as local Markdown.
/mnt/data/MACRO_MANIFEST.md	ASU macro manifest	Module 2 scope, source hierarchy, OLG notation, required derivations, exercise standards, and quality gates.	Read as local Markdown.

1.2 Primary OLG lecture-note source folder

Source	Type	Topics and use	Extraction notes
https://github.com/Bonorinoa/Quals-KB/tree/main/Macroeconomics/OLG	GitHub folder	Inventory shows Solow/growth PDFs, Gustavo homeworks, government/debt slides, and nested OLG notes.	Directory accessible through browser; not all PDFs extracted.
https://raw.githubusercontent.com/Bonorinoa/Quals-KB/refs/heads/main/Macroeconomics/OLG/OLG/OG1.tex	OLG notes	Exchange OLG, sequential markets, lifetime budget, loan-market clearing, examples with identical and heterogeneous endowments.	Raw TeX is Mathpix-style but usable.
https://raw.githubusercontent.com/Bonorinoa/Quals-KB/refs/heads/main/Macroeconomics/OLG/OLG/OG2.tex	OLG notes	Production OLG, CRRA/log saving rates, capital laws, BGP, dynamic efficiency, PAYG, heterogeneity, endogenous labor, home production, public infrastructure exercises.	Raw TeX has line-compressed slides and a few conversion typos; equations were checked before use.

Source	Type	Topics and use	Extraction notes
https://raw.githubusercontent.com/Bonorinoa/Quals-KB/refs/heads/main/Macroeconomics/OLG/slides%20--%20class%20of%20September%2016-2.pdf.tex	Lecture slides	Government in OG economy, taxes/-transfers, government budget, public debt, intergenerational transfers.	Raw TeX is repetitive; used for formulas and topic inventory.
https://raw.githubusercontent.com/Bonorinoa/Quals-KB/refs/heads/main/Macroeconomics/OLG/slides%20--%20class%20of%20September%2019.pdf.tex	Lecture slides	Ricardian equivalence remarks, production with labor supply, intratemporal/intertemporal FOCs, growth example.	Raw TeX is repetitive; used for labor-supply and growth intuition.
Macroeconomics/OLG/GrowthFacts-2025.pdf and Solowclassnotes-2025.pdf	Listed PDFs	Solow and growth-fact background for capital-labor laws and BGP language.	Listed in GitHub folder; not directly extracted in this run.
Macroeconomics/OLG/hw1--hw5-gustavo.pdf	Listed homeworks	Practice-style anchors for OLG, growth, heterogeneity, and government variants.	Listed in GitHub folder; not directly extracted in this run.

1.3 Past ASU macro qualifying-exam anchors

Year	Local file	OLG content used	Exam relevance
2016	2016-06_*.tex	Two-period OLG with capital, skilled/unskilled labor, nested labor aggregator, steady-state return, skill-premium claim.	High.
2017	2017-06_*.tex	OLG with labor-income tax funding public infrastructure; household FOCs, government budget, steady-state capital-labor ratio, tax claim.	High.

Year	Local file	OLG content used	Exam relevance
2018	2018-06_*.tex	Three-period OLG with youth/middle/old, survival, aggregate savings, law of motion, growth of capital-labor ratio.	High.
2019	2019-06_*.tex	Two-period OLG with college/non-college labor profiles, old-age labor income, borrowing, law of motion, dynamic-efficiency claim.	High.
2020	2020-06_*.tex	Two-period OLG with PAYG Social Security, pension formula, steady-state capital-labor ratio, contribution-rate comparative static.	High.
2021	2021-06_*.tex	Two-person household participation threshold, secondary earners, aggregate labor, steady-state output/participation/rental-rate claims.	High.
2022	2022-06_*.tex	Home production, heterogeneous home-good taste, market hours, capital-labor law, GDP-per-worker and return claims.	High.
2023	2023-06_*.tex	Endowment OLG with two types, interest-rate clearing, no-trade case, endowment-loss correlation.	High.
2024	2024-06_*.tex	OLG with market good and non-tradable home good, BGP with market hours, home-productivity shock, Economist GV claim.	High.

1.4 Current-course adjacent anchors

The current 2026 problem sets are not OLG-specific, but they shape current exam style. PS4 emphasizes explicit equilibrium definitions, skilled/unskilled labor, home production, tradable versus non-tradable goods, and Economist X efficiency claims. This is relevant because recent OLG questions mutate the same ingredients inside two-period life-cycle environments.

2 Pattern Analysis

2.1 Core problem archetypes

- A1. Endowment OLG interest-rate clearing.** Solve a lifetime budget, derive a log savings/borrowing rule, aggregate net lending to zero, and interpret the equilibrium gross interest rate.
- A2. Two-period production OLG.** Solve household saving, aggregate young savings into next period capital, normalize by next period effective labor, characterize steady state and BGP.
- A3. Dynamic efficiency/golden rule.** Compare the steady-state gross return on capital to the gross growth rate of the economy.

- A4. Government and PAYG.** Put taxes and pensions into the lifetime budget; the pension is old-age income and crowds out private saving.
- A5. Heterogeneity at birth.** Allow types to differ in labor efficiency, age labor profile, home-good taste, or participation cost; aggregate type-specific savings correctly.
- A6. Home production and participation.** Add time allocation or discrete work choices; derive a threshold or constant-hours condition; then feed labor supply into the capital law.
- A7. Public infrastructure.** Taxes reduce private saving but finance a productive public input; steady-state private capital is generally non-monotone in the tax rate.
- A8. Transitional dynamics.** Capital chosen last period is predetermined today; shocks shift prices immediately only if they affect current productivity or current labor, and shift capital with a one-period lag through saving.

2.2 Recurring exam commands

Past OLG questions repeatedly use the same command verbs:

- formulate the young individual's problem;
- derive FOCs and saving rules;
- define a competitive equilibrium;
- obtain a law of motion for a capital-labor ratio;
- solve or characterize the steady state;
- compare steady-state returns, output per worker, hours, participation, or GDP;
- evaluate an Economist X claim formally and intuitively.

2.3 Recurring state and control objects

The individual state is usually age and type. The aggregate state is usually predetermined capital, sometimes normalized as $k_t = K_t/(A_t L_t)$, K_t/L_t , or K_t/H_t . Controls are consumption, saving, market work, home work, and participation. The market-clearing condition is usually one of these:

$$\begin{array}{ll} \sum_i m_i s_{it} = 0 & \text{endowment-loan economy,} \\ K_{t+1} = L_t s_t & \text{representative young production OLG,} \\ K_{t+1} = \sum_i L_{it} s_{it} & \text{heterogeneous-type production OLG.} \end{array}$$

The most common lost-point pattern

Students often derive the individual saving rule correctly and then divide by the wrong denominator. If s_t is chosen by the young at t , it becomes K_{t+1} . Therefore the normalized law divides by next period's effective labor, $A_{t+1} L_{t+1}$, not by $A_t L_t$.

3 Detailed Blueprint

Element	Blueprint
Module objective	Given any ASU-style OLG problem, set up the household problem, derive the savings rule, aggregate savings, choose the correct normalization, characterize steady state/BGP, and evaluate comparative-static claims.
Prerequisites	Static constrained optimization, log and CRRA utility, Cobb-Douglas factor prices, Solow-style normalization, gross versus net returns.
Canonical models	Endowment OLG; production OLG with log utility; production OLG with CRRA utility; three-period OLG; dynamic efficiency/golden rule; PAYG Social Security; public infrastructure; skilled/unskilled labor; college/non-college age profiles; participation thresholds; home production and non-tradable home goods.
Exam variations	Old-age endowments, pensions, government debt, population growth, labor-augmenting growth, heterogeneous efficiency units, household public consumption, home-good taste heterogeneity, female/secondary-earner productivity, temporary/permanent TFP or home-productivity shocks.
Closed-form catalog	General two-period log savings rule; endowment interest-rate clearing; CRRA saving rate; Cobb-Douglas OLG law and steady state; golden-rule return comparison; PAYG pension and steady-state k ; three-period consumption/saving profile; participation threshold; home-production hours; infrastructure steady state.
Exercise plan	20 drills: mechanical setup, closed-form derivations, past-qual applications, and Economist X traps. Each has a concise answer sketch.
Study use	Block 1: Sections 7–8. Block 2: Sections 11–14. Final review: Section 16.

4 Module Map

Subtopic	Why it matters	Exam appearances	Core technique	Common trap
Endowment OLG	Purest interest-rate clearing logic.	2023 Q1; OG1 examples.	Sum net savings to zero.	Thinking no trade means R is arbitrary.
Two-period production OLG	Backbone for nearly all OLG capital questions.	2016, 2019, 2020, 2021, 2022, 2024.	Savings $\rightarrow K' \rightarrow$ normalize.	Dividing by current labor.

Subtopic	Why it matters	Exam appearances	Core technique	Common trap
CRRA saving	Determines whether return changes raise or lower saving.	OG2 mechanics; possible mutation.	Derive $\mathcal{S}(R)$.	Forgetting R' enters implicitly through k' .
Dynamic efficiency	Used in policy and “Economist” claims.	2019 Q2; OG2 golden rule.	Compare R^* to growth.	Calling all OLG inefficient.
PAYG Social Security	Classic policy application.	2020 Q3; OG2 pensions.	Pension is old-age income.	Missing population growth in p .
Three-period OLG	Tests age indexing and aggregation.	2018 Q2.	Age profiles, cohort masses.	Using one-period formula blindly.
Skilled/unskilled labor	Separates total effects from normalized effects.	2016 Q2; PS4 style.	Normalize by labor aggregator.	Assuming higher efficiency raises return.
College/non-college profiles	Old-age labor income can induce borrowing.	2019 Q2.	Use general old-income savings rule.	Dropping old labor income.
Participation threshold	Discrete labor choice with log closed form.	2021 Q1.	Compare indirect utilities.	Treating threshold as price-dependent when prices scale both options.
Home production	Current and recent high-yield mutation.	2022 Q2; 2024 Q2; PS4 home good.	Time FOC plus savings rule.	Confusing market hours with home hours.
Public infrastructure	Tax has two opposing effects.	2017 Q3; OG2 exercise.	Joint private/public laws.	Claiming higher tax always raises private K .
Transition dynamics	Distinguishes jumps from lags.	OG2 mechanics; 2024 Q2.	Predetermined K_t , shifted law.	Making K_t jump at shock date.

5 Learning Objectives

After studying this module, you should be able to:

1. Given any two-period log OLG problem with old-age income, derive the individual saving rule and identify whether the agent saves or borrows.
2. Given a production OLG environment, derive the law of motion for k_{t+1} with the correct denominator.
3. Given Cobb-Douglas production and log utility, solve for the steady-state normalized capital stock and return.
4. Given a PAYG Social Security system, derive the pension formula, saving rule, and effect of the contribution rate on steady-state capital.
5. Given a type-heterogeneous OLG problem, aggregate savings using type masses and the correct type-specific earnings.
6. Given a home-production or participation problem, derive the time-allocation FOC or threshold and explain its effect on aggregate labor and capital.
7. Given a policy or technology shock, describe the impact, transition, and steady-state effects on wages, returns, capital, output, and hours.
8. Given an Economist X claim, identify the exact formal comparison needed and give the economic intuition.

6 Notation and Timing

Default timing

At date t , the old own K_t , firms rent K_t and labor, young earn labor income, young choose saving s_t , and aggregate young saving becomes next period capital K_{t+1} . A young individual born at t consumes c_{1t} when young and $c_{2,t+1}$ when old.

Unless a problem states otherwise:

$$\begin{aligned}
 L_{t+1} &= (1+n)L_t, & A_{t+1} &= (1+g)A_t, \\
 k_t &= \frac{K_t}{A_t L_t}, & Y_t &= F(K_t, A_t L_t) = A_t L_t f(k_t), \\
 w_t^e &= f(k_t) - k_t f'(k_t), & W_t &= A_t w_t^e, \\
 R_{t+1} &= 1 + r_{t+1} = 1 - \delta + f'(k_{t+1}).
 \end{aligned}$$

Here w_t^e is the wage per efficiency unit and W_t is the wage per young worker with one unit of raw time. If the exam defines $k_t = K_t/L_t$ instead, do not force effective-labor notation; follow the problem and state your normalization.

Gross versus net return

The household Euler equation uses the gross return $R_{t+1} = 1 + r_{t+1}$. Dynamic efficiency is usually easiest in gross terms: compare R^* to the gross growth factor $(1+n)(1+g)$. In net

terms, compare r^* to $(1+n)(1+g) - 1$.

7 Core Derivation Templates

7.1 Template 1: Lifetime budget to log saving rule

Use this whenever the young have current income e_{1t} and old-age non-asset income $e_{2,t+1}$.

$$\begin{aligned} c_{1t} + s_t &= e_{1t}, \\ c_{2,t+1} &= R_{t+1}s_t + e_{2,t+1}. \end{aligned}$$

Combine into the lifetime budget:

$$c_{1t} + \frac{c_{2,t+1}}{R_{t+1}} = e_{1t} + \frac{e_{2,t+1}}{R_{t+1}}.$$

For $u = \log c_{1t} + \beta \log c_{2,t+1}$, the FOC is

$$\frac{1}{c_{1t}} = \beta R_{t+1} \frac{1}{c_{2,t+1}}, \quad c_{2,t+1} = \beta R_{t+1} c_{1t}.$$

Substitution gives

$$\begin{aligned} c_{1t} &= \frac{1}{1+\beta} \left(e_{1t} + \frac{e_{2,t+1}}{R_{t+1}} \right), \\ c_{2,t+1} &= \frac{\beta}{1+\beta} (R_{t+1}e_{1t} + e_{2,t+1}), \\ s_t &= \frac{\beta}{1+\beta} e_{1t} - \frac{1}{1+\beta} \frac{e_{2,t+1}}{R_{t+1}}. \end{aligned}$$

Exam use

This one formula solves old-age endowments, pensions, college old-age labor income, debt transfers, and any two-period log OLG with non-asset old-age income.

7.2 Template 2: Savings aggregation to capital law

1. Solve individual saving s_{it} for each type i .
2. Aggregate: $K_{t+1} = \sum_i L_{it}s_{it}$.
3. Define the capital ratio the problem wants. Common choices:

$$k_{t+1} = \frac{K_{t+1}}{L_{t+1}}, \quad \hat{k}_{t+1} = \frac{K_{t+1}}{A_{t+1}L_{t+1}}, \quad k_{t+1}^H = \frac{K_{t+1}}{H_{t+1}}.$$

4. Divide by the next-period denominator.
5. Substitute competitive factor prices from period t for labor income and from period $t+1$ for returns if returns enter saving.

7.3 Template 3: Steady state versus BGP

A steady state means the normalized state is constant. A balanced growth path means aggregate variables grow at constant rates while normalized ratios are constant. With labor-augmenting growth,

$$k_t = \frac{K_t}{A_t L_t} = k^* \implies \frac{K_{t+1}}{K_t} = (1+n)(1+g), \quad \frac{K_{t+1}/L_{t+1}}{K_t/L_t} = 1+g.$$

Do not call K_t/L_t constant when $g > 0$ unless the problem uses Hicks-neutral productivity or has no labor-augmenting growth.

7.4 Template 4: Comparative statics

For an OLG law $k_{t+1} = \psi(k_t; \theta)$:

1. Identify whether K_t or k_t is predetermined at the shock date.
2. Determine whether the shock changes current prices directly or only future saving.
3. Determine whether the law ψ shifts up or down.
4. Compare new and old steady states.
5. Use $w(k)$ increasing and $R(k)$ decreasing under Cobb-Douglas.

8 Canonical Models and Derivations

8.1 Model 1: Two-period endowment OLG

Environment. Each individual lives for two periods. Type i has endowment e_{1i} when young and e_{2i} when old. Cohort masses are m_i . There is a one-period loan market with gross return R_t .

Individual problem.

$$\max_{c_{1i}, c_{2i}, s_i} \log c_{1i} + \beta \log c_{2i}$$

subject to

$$\begin{aligned} c_{1i} + s_i &= e_{1i}, \\ c_{2i} &= R s_i + e_{2i}. \end{aligned}$$

The log rule from Template 1 gives

$$s_i(R) = \frac{\beta}{1+\beta} e_{1i} - \frac{1}{1+\beta} \frac{e_{2i}}{R}.$$

Equilibrium. Loans are in zero net supply:

$$\sum_i m_i s_i(R) = 0.$$

Therefore

$$R^* = \frac{\sum_i m_i e_{2i}}{\beta \sum_i m_i e_{1i}}.$$

Intuition. If old-age endowments are high relative to young-age endowments, agents want to borrow when young, so the equilibrium interest rate must be high enough to discourage borrowing. If young-age endowments are high, the interest rate can be lower.

2023-style old-age loss. Suppose type $i \in \{A, B\}$ has $e_{1i} = z_i$ and $e_{2i} = z_i(1 - \delta_i)$, with equal masses. Then

$$R^* = \frac{z_A(1 - \delta_A) + z_B(1 - \delta_B)}{\beta(z_A + z_B)} < \frac{1}{\beta}.$$

If $\delta_A = \delta_B = \bar{\delta}$, then $R^* = (1 - \bar{\delta})/\beta$ and

$$s_i = \frac{\beta z_i}{1 + \beta} - \frac{z_i(1 - \bar{\delta})}{(1 + \beta)R^*} = 0,$$

so there is no trade. If $z_A > z_B$ and high young endowment is associated with a larger old-age loss, then aggregate old-age endowment falls relative to aggregate young-age endowment, so R^* is lower than in the common-loss case.

No trade does not mean no price

In an endowment OLG with identical age profiles, equilibrium allocations may be autarkic, but the interest rate is still pinned down by the condition that autarky is individually optimal.

8.2 Model 2: Two-period production OLG with log utility

Environment. Young individuals supply one unit of labor, earn $W_t = A_t w_t^e$, save s_t , and consume old-age return $R_{t+1}s_t$. Firms operate

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad 0 < \alpha < 1.$$

Let $k_t = K_t/(A_t L_t)$.

Prices.

$$\begin{aligned} w_t^e &= (1 - \alpha)k_t^\alpha, \\ W_t &= A_t(1 - \alpha)k_t^\alpha, \\ R_{t+1} &= 1 - \delta + \alpha k_{t+1}^{\alpha-1}. \end{aligned}$$

Household problem.

$$\begin{aligned} \max_{c_{1t}, c_{2,t+1}, s_t} \quad & \log c_{1t} + \beta \log c_{2,t+1} \\ \text{s.t.} \quad & c_{1t} + s_t = W_t, \\ & c_{2,t+1} = R_{t+1}s_t. \end{aligned}$$

The log rule with no old-age non-asset income is

$$s_t = \frac{\beta}{1 + \beta} W_t.$$

Aggregation and normalization. Market clearing for capital is

$$K_{t+1} = L_t s_t = L_t \frac{\beta}{1+\beta} A_t (1-\alpha) k_t^\alpha.$$

Divide by $A_{t+1} L_{t+1} = (1+g)(1+n)A_t L_t$:

$$k_{t+1} = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} k_t^\alpha.$$

Steady state. A steady state in effective units solves

$$k^* = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} (k^*)^\alpha.$$

Thus

$$k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}.$$

BGP implications. Along the BGP, k_t is constant, but K_t , Y_t , and aggregate consumption grow at $(1+n)(1+g)$. Per-worker capital and output grow at $1+g$. The wage per efficiency unit is constant; the wage per worker grows at $1+g$. The gross return is constant.

8.3 Model 3: CRRA utility and the implicit OLG law

Let

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0,$$

with the log case obtained as $\sigma \rightarrow 1$. The household problem is

$$\begin{aligned} c_{1t} + s_t &= W_t, \\ c_{2,t+1} &= R_{t+1} s_t. \end{aligned}$$

The Euler equation is

$$c_{1t}^{-\sigma} = \beta R_{t+1} c_{2,t+1}^{-\sigma}.$$

Using $c_{2,t+1} = R_{t+1} s_t$ and $c_{1t} = W_t - s_t$,

$$\left(\frac{s_t}{W_t - s_t} \right)^\sigma = \beta R_{t+1}^{1-\sigma}.$$

Therefore

$$s_t = \mathcal{S}(R_{t+1}) W_t, \quad \mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}}.$$

The equilibrium law becomes implicit because R_{t+1} depends on k_{t+1} :

$$(1+n)(1+g)k_{t+1} = \mathcal{S}(1-\delta + f'(k_{t+1})) [f(k_t) - k_t f'(k_t)].$$

Return effect on saving

If $\sigma = 1$, the saving rate is $\beta/(1 + \beta)$ and is independent of R . If $\sigma < 1$, substitution effects dominate and saving rises with R . If $\sigma > 1$, income effects dominate and saving falls with R .

8.4 Model 4: Dynamic efficiency and the golden rule

Dynamic efficiency is not the same as “capital is useful” or “taxes are absent.” It is a comparison between the marginal product of installed capital and the growth rate of the economy.

Let output per efficiency unit be $f(k)$, with depreciation δ and growth in efficiency units equal to $1 + n + g$. Aggregate feasibility in effective units is

$$\frac{C_t}{A_t L_t} + (1 + n)(1 + g)k_{t+1} = f(k_t) + (1 - \delta)k_t.$$

If the economy holds a constant effective capital stock k , maximum sustainable consumption per efficiency unit is

$$\bar{C}(k) = f(k) + (1 - \delta)k - (1 + n)(1 + g)k.$$

The golden-rule capital stock solves

$$\begin{aligned}\bar{C}'(k_{GR}) &= 0, \\ f'(k_{GR}) + 1 - \delta &= (1 + n)(1 + g).\end{aligned}$$

Thus the golden-rule condition can be written as

$$R(k_{GR}) = (1 + n)(1 + g), \quad R(k) = 1 - \delta + f'(k).$$

Dynamic-efficiency test

A steady state k^* is dynamically inefficient if $k^* > k_{GR}$, equivalently if

$$R(k^*) < (1 + n)(1 + g).$$

It is dynamically efficient if $R(k^*) \geq (1 + n)(1 + g)$. In words: if the economy’s gross return is below the gross growth factor, reducing capital can raise consumption for all generations after a transition.

For Cobb-Douglas production with full depreciation,

$$R^* = \alpha(k^*)^{\alpha-1}.$$

Using the log-utility steady state from Model 2,

$$(k^*)^{1-\alpha} = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)},$$

so

$$R^* = \frac{\alpha(1+\beta)(1+n)(1+g)}{\beta(1-\alpha)}.$$

Therefore this benchmark is dynamically efficient if and only if

$$\frac{\alpha(1+\beta)}{\beta(1-\alpha)} \geq 1.$$

A higher saving motive β can push the economy toward overaccumulation. A higher capital share α raises the marginal product of capital and pushes toward efficiency.

Qualifier trap

Do not test dynamic inefficiency by asking whether k^* rises or falls after a policy. Always compare the equilibrium return to the economy's growth factor. In no-growth, full-depreciation models this becomes $R^* < 1$; with population or productivity growth it becomes $R^* < (1+n)(1+g)$.

8.5 Model 5: Pay-as-you-go Social Security

This is the 2020-style OLG mutation. Young agents work, pay a payroll tax, save privately, and receive a pension when old.

Environment. Young population grows as $N_{t+1}^y = (1+n)N_t^y$. Preferences are

$$\log c_{1t} + \beta \log c_{2,t+1}.$$

The government taxes a fraction τ of young labor income and pays a pension p_t to each old agent alive at date t .

Government budget. At date t , current young contributions finance current old benefits:

$$N_t^y \tau w_t = N_{t-1}^y p_t.$$

Since $N_t^y = (1+n)N_{t-1}^y$,

$$p_t = (1+n)\tau w_t.$$

A generation born at date t therefore expects old-age pension

$$p_{t+1} = (1+n)\tau w_{t+1}.$$

Household problem. The household solves

$$\begin{aligned} \max_{c_{1t}, c_{2,t+1}, s_t} \quad & \log c_{1t} + \beta \log c_{2,t+1} \\ \text{s.t.} \quad & c_{1t} + s_t = (1-\tau)w_t, \\ & c_{2,t+1} = R_{t+1}s_t + p_{t+1}. \end{aligned}$$

Use the generic old-age-income formula with $e_{1t} = (1-\tau)w_t$ and $e_{2,t+1} = p_{t+1}$:

$$s_t = \frac{\beta}{1+\beta}(1-\tau)w_t - \frac{1}{1+\beta} \frac{p_{t+1}}{R_{t+1}}.$$

Substituting the PAYG pension formula,

$$s_t = \frac{\beta}{1+\beta}(1-\tau)w_t - \frac{1}{1+\beta} \frac{(1+n)\tau w_{t+1}}{R_{t+1}}.$$

Closed-form capital law under Cobb-Douglas and full depreciation. Let $f(k) = k^\alpha$, $w_t = (1 - \alpha)k_t^\alpha$, and $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$. Then

$$\frac{w_{t+1}}{R_{t+1}} = \frac{(1 - \alpha)k_{t+1}^\alpha}{\alpha k_{t+1}^{\alpha-1}} = \frac{1 - \alpha}{\alpha} k_{t+1}.$$

Capital market clearing is $(1 + n)k_{t+1} = s_t$. Therefore

$$(1 + n)k_{t+1} = \frac{\beta(1 - \tau)(1 - \alpha)}{1 + \beta} k_t^\alpha - \frac{(1 + n)\tau}{1 + \beta} \frac{1 - \alpha}{\alpha} k_{t+1},$$

$$(1 + n)k_{t+1} \left[(1 + \beta) + \tau \frac{1 - \alpha}{\alpha} \right] = \beta(1 - \tau)(1 - \alpha)k_t^\alpha.$$

Thus

$$k_{t+1} = \frac{\beta(1 - \tau)(1 - \alpha)}{(1 + n) \left[(1 + \beta) + \tau \frac{1 - \alpha}{\alpha} \right]} k_t^\alpha.$$

The steady state is

$$k^*(\tau) = \left[\frac{\beta(1 - \tau)(1 - \alpha)}{(1 + n) \left[(1 + \beta) + \tau \frac{1 - \alpha}{\alpha} \right]} \right]^{\frac{1}{1 - \alpha}}.$$

Because the numerator falls and the denominator rises with τ , $k^*(\tau)$ is strictly decreasing in τ .

Economic intuition. PAYG Social Security crowds out private saving through two channels. First, the payroll tax lowers disposable young income. Second, the pension is old-age income, so the young need less private wealth to finance old-age consumption. The implicit gross return on PAYG is approximately the growth factor of the tax base, $(1 + n)(1 + g)$ when wages grow at $1 + g$. If the market return is larger, PAYG tends to lower lifetime resources for young generations; if the market return is lower, PAYG can help with dynamic inefficiency.

8.6 Model 6: Three-period OLG

The 2018-style three-period problem is a setup trap. The main difficulty is not calculus; it is dating each cohort's wage, debt, and savings correctly.

Environment. Agents live as young, middle-aged, and old. They receive labor income only when middle-aged. Preferences of a generation born at date t are

$$\log c_t^y + \beta \log c_{t+1}^m + \beta^2 \log c_{t+2}^o.$$

Let

$$D = 1 + \beta + \beta^2.$$

The lifetime budget is

$$c_t^y + \frac{c_{t+1}^m}{R_{t+1}} + \frac{c_{t+2}^o}{R_{t+1}R_{t+2}} \leq \frac{w_{t+1}}{R_{t+1}}.$$

The middle-aged earn w_{t+1} at date $t + 1$, so young agents generally borrow.

FOCs and consumption profile. The first-order conditions imply

$$c_{t+1}^m = \beta R_{t+1} c_t^y, \quad c_{t+2}^o = \beta R_{t+2} c_{t+1}^m.$$

Substitute into the lifetime budget:

$$c_t^y + \beta c_t^y + \beta^2 c_t^y = \frac{w_{t+1}}{R_{t+1}}.$$

Therefore

$$\begin{aligned} c_t^y &= \frac{w_{t+1}}{R_{t+1} D}, \\ c_{t+1}^m &= \frac{\beta w_{t+1}}{D}, \\ c_{t+2}^o &= \frac{\beta^2 R_{t+2} w_{t+1}}{D}. \end{aligned}$$

Individual asset positions. Young agents save

$$s_t^y = -c_t^y = -\frac{w_{t+1}}{R_{t+1} D}.$$

Middle-aged agents save for old age:

$$s_{t+1}^m = w_{t+1} + R_{t+1} s_t^y - c_{t+1}^m = \frac{\beta^2 w_{t+1}}{D}.$$

Old agents dissave all remaining assets.

Aggregate law. At date t , aggregate next-period capital is

$$K_{t+1} = N_t^y s_t^y + N_t^m s_t^m.$$

In the 2018 formulation, the survival/discount parameter implies $N_t^m = \beta N_t^y$, so $N_t^y/N_t^m = 1/\beta$. Let labor be supplied by the middle-aged, $L_t = N_t^m$, and let $k_t = K_t/L_t$.

With full depreciation and Cobb-Douglas production $Y_t = A_t K_t^\alpha L_t^{1-\alpha}$,

$$w_t = (1 - \alpha) A_t k_t^\alpha, \quad R_{t+1} = \alpha A_{t+1} k_{t+1}^{\alpha-1}.$$

Thus $w_{t+1}/R_{t+1} = ((1 - \alpha)/\alpha) k_{t+1}$. Divide aggregate capital by $L_t = N_t^m$:

$$\begin{aligned} k_{t+1} &= \frac{\beta^2 (1 - \alpha) A_t k_t^\alpha}{D} - \frac{1}{\beta D} \frac{1 - \alpha}{\alpha} k_{t+1}, \\ k_{t+1} &= \frac{\beta^2 (1 - \alpha)}{D + \frac{1 - \alpha}{\alpha \beta}} A_t k_t^\alpha. \end{aligned}$$

Balanced growth. Because A_t is Hicks-neutral in this formulation, a BGP for K_t/L_t has growth factor $(A_{t+1}/A_t)^{1/(1-\alpha)}$. This differs from the labor-augmenting convention in Model 2, where $K/(AL)$ is constant and K/L grows at $1 + g$. Always identify whether technology is Hicks-neutral, $A_t K^\alpha L^{1-\alpha}$, or labor-augmenting, $F(K, A_t L)$.

Dating trap

In the three-period model, the young do not earn the wage w_t at birth. They borrow against w_{t+1} . Missing this changes the sign of young saving and destroys the aggregate capital law.

8.7 Model 7: Heterogeneous labor profiles

Past quals repeatedly add heterogeneity in labor efficiency or timing. The robust workflow is: solve type-specific savings, aggregate in dollars, then normalize by the labor aggregate used by firms.

8.7.1 Skilled and unskilled labor

Suppose production is

$$Y_t = K_t^\alpha H_t^{1-\alpha}, \quad H_t = S_t^\phi U_t^{1-\phi},$$

where S_t and U_t are efficiency units of skilled and unskilled labor. With log utility and no old-age non-asset income, each young agent saves the constant fraction $\beta/(1+\beta)$ of labor income. Hence aggregate saving is

$$K_{t+1} = \frac{\beta}{1+\beta} w_t^H H_t = \frac{\beta}{1+\beta} (1-\alpha) Y_t.$$

Let $k_t = K_t/H_t$. If $H_{t+1}/H_t = (1+n)(1+g_H)$, then

$$(1+n)(1+g_H)k_{t+1} = \frac{\beta}{1+\beta} (1-\alpha) k_t^\alpha.$$

Thus a permanent increase in skilled efficiency can raise output levels and wages but may leave normalized k^* and the rental rate unchanged if it scales effective labor rather than changing saving rates.

For the nested Cobb-Douglas labor aggregate, relative factor prices satisfy

$$\frac{w_S}{w_U} = \frac{\phi}{1-\phi} \frac{U_t}{S_t}.$$

If $S_t = z_s N_s$ and $U_t = z_u N_u$, the earnings premium per worker is

$$\frac{w_S z_s}{w_U z_u} = \frac{\phi}{1-\phi} \frac{N_u}{N_s}.$$

In this particular aggregator, the efficiency terms cancel in the worker-level skill premium. That is exactly why a claim such as “higher skilled productivity must raise the skill premium” must be checked against the labor aggregator.

8.7.2 College and non-college labor profiles

Let type $i \in \{C, NC\}$ have young labor efficiency a_i and old labor efficiency b_i . If the type- i young agent born at t earns $w_t a_i$ when young and $w_{t+1} b_i$ when old, then

$$\begin{aligned} c_{1it} + s_{it} &= w_t a_i, \\ c_{2i,t+1} &= R_{t+1} s_{it} + w_{t+1} b_i. \end{aligned}$$

The log savings rule is

$$s_{it} = \frac{\beta}{1+\beta} w_t a_i - \frac{1}{1+\beta} \frac{w_{t+1} b_i}{R_{t+1}}.$$

Backloaded earnings profiles can imply borrowing when young. Aggregate capital is

$$K_{t+1} = N_t \sum_i \pi_i s_{it}.$$

The capital law is implicit because w_{t+1}/R_{t+1} is proportional to k_{t+1} in Cobb-Douglas models. Increasing the fraction of a backloaded type can reduce current saving even if that type has high lifetime earnings.

Composition trap

Do not conclude that a larger share of high-lifetime-income households raises steady-state capital. Check whether income is frontloaded or backloaded. Young saving depends positively on young income and negatively on discounted old income.

8.8 Model 8: Participation thresholds and endogenous market hours

8.8.1 Continuous hours

Let the young choose market hours h_t and leisure $1 - h_t$ with utility

$$\log c_{1t} + B \frac{(1 - h_t)^{1-\sigma}}{1 - \sigma} + \beta \log c_{2,t+1}.$$

The budget constraints are

$$c_{1t} + s_t = W_t h_t, \quad c_{2,t+1} = R_{t+1} s_t.$$

The Euler equation gives $s_t = \beta c_{1t}$, so

$$c_{1t} = \frac{W_t h_t}{1 + \beta}, \quad s_t = \frac{\beta W_t h_t}{1 + \beta}.$$

The intratemporal FOC is

$$\frac{W_t}{c_{1t}} = B(1 - h_t)^{-\sigma}.$$

Substitute $c_{1t} = W_t h_t / (1 + \beta)$:

$$\frac{1 + \beta}{h_t} = B(1 - h_t)^{-\sigma}.$$

Hours are constant because income and substitution effects cancel under log market consumption and this separable leisure form.

8.8.2 Participation thresholds

The 2021-style threshold question uses a fixed cost or disutility q of having both household members work. Let one member have efficiency 1 and the potential secondary earner have efficiency h . If both work, effective labor income is proportional to $1 + h$; if only one works, it is proportional to 1.

With log utility for two people in each period, the indirect value gain from having the second earner is

$$2(1 + \beta) \log(1 + h),$$

after dropping terms common across participation choices. The household chooses two workers if

$$q \leq q^*(h) = 2(1 + \beta) \log(1 + h).$$

Since

$$\frac{\partial q^*}{\partial h} = \frac{2(1 + \beta)}{1 + h} > 0,$$

higher secondary-earner productivity raises the participation threshold. If $G(q)$ is the distribution of participation costs, the fraction of two-earner households is $G(q^*)$.

If primary workers have efficiency 1 and secondary workers have efficiency h , aggregate effective labor is

$$H_t = N_t [1 + hG(q^*)].$$

In Cobb-Douglas models with log saving, normalized capital per effective worker may converge to the same k^* even though total output, raw output per household, and participation differ.

Participation trap

A higher participation rate raises aggregate labor and output. It does not automatically lower the steady-state rental rate if the capital law is written in capital per effective worker and the saving rate is unchanged.

8.9 Model 9: Home production and non-tradable home goods

8.9.1 Home production with market work only when young

This is the 2022-style version. Agents work in the market only when young, but can produce home goods in both periods. Home goods are perishable and cannot be stored. Let young home time be l_t , market time be $m_t = 1 - l_t$, and home technology be

$$h_t = G(l_t) = l_t^\lambda, \quad \lambda \in (0, 1).$$

Old agents do not work in the market, so old home time is optimally 1.

The young household of type γ solves

$$\begin{aligned} \max_{c_1, c_2, s, l} \quad & \log c_1 + \gamma \log(l^\lambda) + \beta [\log c_2 + \gamma \log(1^\lambda)] \\ \text{s.t.} \quad & c_1 + s = w(1 - l), \\ & c_2 = Rs. \end{aligned}$$

The Euler equation gives $s = \beta c_1$. Hence

$$c_1 = \frac{w(1 - l)}{1 + \beta}, \quad s = \frac{\beta w(1 - l)}{1 + \beta}.$$

The home-time FOC is

$$\frac{w}{c_1} = \frac{\gamma \lambda}{l}.$$

Substitute $c_1 = w(1 - l)/(1 + \beta)$:

$$\frac{1 + \beta}{1 - l} = \frac{\gamma \lambda}{l}.$$

Therefore

$$l(\gamma) = \frac{\gamma\lambda}{1 + \beta + \gamma\lambda}, \quad m(\gamma) = 1 - l(\gamma) = \frac{1 + \beta}{1 + \beta + \gamma\lambda},$$

and

$$s(\gamma) = \frac{\beta w}{1 + \beta + \gamma\lambda}.$$

A higher taste for home goods lowers market hours and private saving.

With two types L and H , $\gamma_L < \gamma_H$, and type- L share π , aggregate market labor is

$$M = N [\pi m(\gamma_L) + (1 - \pi)m(\gamma_H)].$$

Aggregate saving is

$$\begin{aligned} K' &= N [\pi s(\gamma_L) + (1 - \pi)s(\gamma_H)] \\ &= N \frac{\beta}{1 + \beta} w [\pi m(\gamma_L) + (1 - \pi)m(\gamma_H)] \\ &= \frac{\beta}{1 + \beta} w M. \end{aligned}$$

Thus, if $k = K/M$ and $Y = K^\alpha M^{1-\alpha}$,

$$k' = \frac{\beta}{1 + \beta} (1 - \alpha) k^\alpha.$$

The normalized steady-state return is independent of type composition, but GDP per household is higher when the economy has more low- γ households because market hours are higher.

8.9.2 Non-tradable home good and BGP hours

The 2024-style model adds population growth, labor-efficiency growth, and a non-tradable home good. Let market-good utility be log and home-good utility be isoelastic:

$$v(c_1, c_2) = \log c_1 + \frac{c_2^{1-\sigma}}{1-\sigma}.$$

Home production is linear:

$$c_{2t} = A h_t, \quad m_t + h_t = 1.$$

Let W_t denote the market wage per unit of market time. The intratemporal FOC is

$$\frac{W_t}{c_{1t}} = A(A h_t)^{-\sigma} = A^{1-\sigma} h_t^{-\sigma}.$$

This condition is the key to BGP reasoning. Along a BGP with fixed A , log market utility often makes W_t/c_{1t} stationary, so h_t and m_t can be constant. But after a permanent change in home productivity A , the right-hand side changes unless h_t adjusts.

Economist GV claim test

The claim “market hours remain constant after a permanent increase in home productivity because the economy returns to a BGP” is generally false. BGP means hours are constant along the new path, not that they equal their old-BGP value. The new constant hours must

satisfy the new intratemporal FOC.

Capital-law logic. Let market production be

$$Y_t = K_t^\alpha (X_t N_t m_t)^{1-\alpha}, \quad X_{t+1} = (1+g)X_t, \quad N_{t+1} = (1+n)N_t.$$

Define

$$k_t = \frac{K_t}{X_t N_t m_t}.$$

When m_t is constant on a BGP, the standard log-saving law applies in effective market units. If a shock changes m_t on impact, K_t is predetermined but $X_t N_t m_t$ is not; therefore measured k_t can jump even though aggregate capital cannot.

8.10 Model 10: Public infrastructure funded by a labor tax

The 2017-style public-infrastructure problem is a policy-comparative-static problem, not just a tax-wedge problem.

Environment. Suppose output per worker is

$$y_t = z_t^\lambda k_t^\alpha, \quad \alpha, \lambda > 0, \quad \alpha + \lambda < 1,$$

where z_t is public infrastructure. Young agents pay labor-income tax τ ; the after-tax wage is $(1-\tau)(1-\alpha)y_t$. With log utility,

$$k_{t+1} = \frac{\beta}{1+\beta} (1-\tau)(1-\alpha) z_t^\lambda k_t^\alpha.$$

Let

$$a = \frac{\beta(1-\alpha)}{1+\beta}.$$

Then

$$k_{t+1} = a(1-\tau) z_t^\lambda k_t^\alpha.$$

Government investment evolves as

$$z_{t+1} = (1-\delta_z)z_t + au(1-\alpha)z_t^\lambda k_t^\alpha.$$

Steady state. At a steady state,

$$\begin{aligned} k &= a(1-\tau)z^\lambda k^\alpha, \\ \delta_z z &= \tau(1-\alpha)z^\lambda k^\alpha. \end{aligned}$$

Using the first equation to substitute $z^\lambda k^\alpha = k/[a(1-\tau)]$ into the second,

$$z^* = \frac{\tau(1+\beta)}{\delta_z \beta(1-\tau)} k^*.$$

Substitute this into the capital equation:

$$k^* = \left[\frac{\beta(1-\alpha)(1-\tau)}{1+\beta} \left(\frac{\tau(1+\beta)}{\delta_z \beta(1-\tau)} \right)^\lambda \right]^{\frac{1}{1-\alpha-\lambda}}.$$

Comparative static. The tax has two opposing effects. It lowers disposable wage income and private saving through $(1 - \tau)$, but it raises infrastructure through τ . In the closed form above, the policy component is proportional to

$$\tau^\lambda(1 - \tau)^{1-\lambda}.$$

This is maximized at $\tau = \lambda$. Therefore a claim that a higher tax always lowers or always raises steady-state capital is false without knowing whether τ is below or above the infrastructure elasticity.

9 Steady State, BGP, Transitional Dynamics, and Comparative Statics

9.1 Steady state versus balanced growth path

A steady state is a constant value of the normalized state. A BGP is a path along which aggregate variables grow at constant rates and normalized variables are constant. In OLG, the normalization is often the entire question.

Technology convention	Correct normalized capital	BGP statement
Labor augmenting: $F(K, AL)$	$k = K/(AL)$	k constant; K/L grows at $1 + g$.
Hicks neutral: $AK^\alpha L^{1-\alpha}$	$\tilde{k} = (K/L)/A^{1/(1-\alpha)}$	K/L grows at $(1 + g_A)^{1/(1-\alpha)}$.
Market hours enter labor: $F(K, ALm)$	$k = K/(ALm)$	If m is constant, standard BGP; if m changes, normalized k can jump.
Public infrastructure: $z^\lambda k^\alpha$	two states, (k, z)	BGP/steady state requires both normalized states to be constant.

9.2 Transition rules that save points

- Rule 1: Capital chosen last period is predetermined.** K_t cannot jump at date t . Prices can jump because they depend on current K_t , current productivity, current labor, and expected future capital.
- Rule 2: Future capital can jump relative to the old law.** K_{t+1} is chosen by date- t young savers, so preference, tax, pension, wage, and participation changes affect it immediately.
- Rule 3: Returns usually fall when capital rises.** In Cobb-Douglas, $R = 1 - \delta + \alpha k^{\alpha-1}$. If a shock raises the long-run k , it lowers the long-run return.
- Rule 4: Wages usually rise when capital rises.** In Cobb-Douglas, $w = (1 - \alpha)Ak^\alpha$ per efficiency unit.
- Rule 5: Composition effects require a denominator.** Ask whether the variable is total, per person, per raw worker, per effective worker, or per market worker.

9.3 Canonical shock responses

Permanent increase in β . In the log two-period benchmark, the capital law shifts up because the saving rate $\beta/(1 + \beta)$ rises. The transition has higher k_{t+1} for a given k_t , a higher new k^* , higher

long-run wages, and lower long-run returns. Initial old agents are affected by the date- t return if the price/return is determined at date t ; young agents change saving immediately.

Permanent increase in TFP. If TFP is labor augmenting and the normalized law is unchanged, the new BGP has higher levels but the same normalized k^* . If TFP enters as a level in $AK^\alpha L^{1-\alpha}$, the capital-labor ratio grows to a new path. Date- t wages can jump because current productivity jumps; date- t capital cannot.

Temporary TFP increase. A temporary wage increase raises current saving of the young. If agents know productivity returns to normal, the future capital stock can rise temporarily and then converge back. Do not draw the permanent-shift diagram for a temporary shock.

Permanent increase in Social Security tax τ . The date- t young save less because disposable income falls and expected pensions rise. The capital law shifts down, lowering k^* and raising the long-run marginal product of capital. Whether welfare rises depends on dynamic efficiency and intergenerational redistribution; the capital comparative static alone is not a welfare answer.

Permanent increase in home productivity A . The home-good FOC changes. Home hours generally change on impact; market hours and the denominator of effective market capital can change even though physical capital is predetermined. The new BGP has constant hours, but not necessarily the old hours.

10 Closed-Form Catalog

Object	Formula	Use
Two-period log saving with old income	$s_t = \frac{\beta}{1+\beta} e_{1t} - \frac{1}{1+\beta} \frac{e_{2,t+1}}{R_{t+1}}$	Universal setup shortcut.
Endowment OLG interest clearing	$R = \frac{\sum_i \pi_i e_{2i}}{\beta \sum_i \pi_i e_{1i}}$	Pure exchange/no capital.
Production OLG log law	$k_{t+1} = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} k_t^\alpha$	Benchmark dynamics.
Production OLG steady state	$k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} \right]^{1/(1-\alpha)}$	Baseline k^* .
CRRA saving rate	$\mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}}$	Return-dependent saving.
Golden-rule return	$R(k_{GR}) = (1+n)(1+g)$	Dynamic efficiency.
PAYG pension	$p_t = (1+n)\tau w_t$	Social Security setup.
PAYG saving	$s_t = \frac{\beta}{1+\beta} (1-\tau) w_t - \frac{1}{1+\beta} \frac{(1+n)\tau w_{t+1}}{R_{t+1}}$	Crowding out.
Three-period young consumption	$c_t^y = w_{t+1}/(R_{t+1}D)$, $D = 1 + \beta + \beta^2$	Three-period dating.

Object	Formula	Use
Participation threshold	$q^*(h) = 2(1 + \beta) \log(1 + h)$	Household participation.
Home time, $G(l) = l^\lambda$	$l(\gamma) = \frac{\gamma\lambda}{1 + \beta + \gamma\lambda}$	Home production.
Public infrastructure steady z/k	$z^*/k^* = \frac{\tau(1 + \beta)}{\delta_z\beta(1 - \tau)}$	Public capital.

11 Exam Variations

Year	Variation	What changes and what stays invariant
2016	Skilled/unskilled OLG	Replace raw labor with an effective labor aggregator. Log saving still aggregates to a constant fraction of labor income. Skill-premium claims depend on the aggregator.
2017	Public infrastructure	State is (k, z) , not just k . Tax has saving cost and productivity benefit. Steady-state tax comparative static is nonmonotone.
2018	Three-period OLG	Young borrow, middle-aged save. Aggregate capital is middle-aged saving minus young borrowing. BGP growth depends on the productivity convention.
2019	College/non-college profiles	Type-specific old labor income enters as negative saving. Backloaded earnings can reduce young saving. Dynamic inefficiency requires return-growth comparison.
2020	Social Security	Pension is old-age income. PAYG budget pins down $p_t = (1 + n)\tau w_t$. Higher τ lowers k^* in the Cobb-Douglas/log benchmark.
2021	Participation threshold	Solve discrete work choice by indirect utility. Higher secondary productivity raises the threshold and participation. Rental-rate claims require normalization.
2022	Home production	Home preference lowers market hours and saving. Type composition affects raw GDP per person but may not affect normalized return.
2023	Endowment OLG	No capital law. Interest rate clears loans. No-trade arises when all types have identical desired saving at autarky.

Year	Variation	What changes and what stays invariant
2024	Non-tradable home good	BGP has constant hours only after solving the new intratemporal FOC. Home-productivity shocks generally change hours and transition dynamics.

12 Common Traps and Grader Rewards

High-frequency traps

1. Confusing $K_{t+1} = N_t s_t$ with $k_{t+1} = s_t$. Divide by next-period labor or effective labor.
2. Using w_t instead of w_{t+1} for old or middle-age income.
3. Forgetting that old-age non-asset income lowers saving.
4. Treating a BGP as a literal steady state of aggregate variables.
5. Testing dynamic efficiency by the sign of saving instead of comparing R^* to the growth factor.
6. Assuming higher productivity raises the skill premium without checking the labor aggregator.
7. Calling non-tradability inefficient without identifying a missing market or wedge relative to the relevant feasible set.
8. Ignoring that market hours can change the denominator of capital per effective market worker.

What graders reward

1. A clean lifetime budget before any FOC.
2. A savings rule with old-age income shown explicitly.
3. Correct aggregation with population and efficiency growth.
4. A clearly labeled normalized state variable.
5. Comparative statics that separate impact, transition, and long-run effects.
6. Formal answer plus one-sentence intuition for every Economist X claim.

13 Exercise Bank

13.1 Mechanical derivation drills

- M1. Skill: lifetime budget.** A young agent has income e_1 , old-age income e_2 , and gross return R . Utility is $\log c_1 + \beta \log c_2$. Derive c_1 , c_2 , and s .

- M2. Skill: aggregate savings.** In a two-period OLG with $N_{t+1} = (1+n)N_t$, show why $k_{t+1} = s_t/(1+n)$ when $k = K/N$ and no efficiency growth.
- M3. Skill: effective-labor normalization.** Suppose production is $F(K, AL)$ and $A_{t+1} = (1+g)A_t$. Derive the law for $k = K/(AL)$ from $K_{t+1} = N_t s_t$.
- M4. Skill: factor prices.** For $Y = K^\alpha(AL)^{1-\alpha}$, derive w_t per worker, W_t per efficiency unit, and R_t with depreciation δ .
- M5. Skill: government budget.** In PAYG Social Security, derive p_t as a function of τ , w_t , and n .

13.2 Closed-form and proof drills

- C1. Skill: Cobb-Douglas steady state.** Derive k^* in the log two-period benchmark with population and efficiency growth.
- C2. Skill: dynamic efficiency.** For the same model, derive the condition under which the steady state is dynamically inefficient.
- C3. Skill: CRRA saving.** Derive the saving rate $S(R)$ and sign its derivative with respect to R for $\sigma < 1$, $\sigma = 1$, and $\sigma > 1$.
- C4. Skill: participation threshold.** Derive $q^*(h)$ when a two-person household pays disutility q if both work.
- C5. Skill: public-infrastructure steady state.** Derive the closed-form $k^*(\tau)$ and identify why the tax effect is nonmonotone.

13.3 Past-qual-style applications

- P1. Skill: three-period aggregation.** In the three-period model, derive the aggregate capital law and explain why young borrowing appears with a minus sign.
- P2. Skill: Social Security comparative static.** Show that a higher payroll tax lowers k^* in the 2020 Cobb-Douglas/log benchmark.
- P3. Skill: home production and type composition.** In the 2022 home production model, compare two economies with different shares of low- γ households. Which has higher GDP per household? Which has lower return?
- P4. Skill: endowment OLG interest clearing.** With two types and endowments $(z_i, z_i(1 - \delta_i))$, derive the equilibrium gross return.
- P5. Skill: BGP home-good shock.** In the 2024 model, explain why a permanent rise in home productivity generally changes market hours even though both old and new paths are BGPs.

13.4 Trap and Economist-claim questions

- T1. Skill: evaluate dynamic-inefficiency claim.** Economist X says: “If college types save less, the economy must be dynamically inefficient.” Evaluate.
- T2. Skill: total versus per-worker effects.** Economist X says: “More secondary-earner participation must lower the rental rate because there is more saving.” Evaluate.

- T3. Skill: nontradability and efficiency.** Economist X says: “If home goods cannot be traded, equilibrium is automatically inefficient.” Evaluate.
- T4. Skill: PAYG return.** Economist X says: “Social Security is good if population growth is positive.” Evaluate formally.
- T5. Skill: TFP transition.** Economist X says: “A permanent TFP increase makes capital jump immediately.” Evaluate.

14 Answer Sketches

14.1 Mechanical drills

M1. Setup:

$$c_1 + s = e_1, \quad c_2 = Rs + e_2.$$

Key equation: $c_2 = \beta Rc_1$. Final result:

$$c_1 = \frac{e_1 + e_2/R}{1 + \beta}, \quad c_2 = \frac{\beta(Re_1 + e_2)}{1 + \beta}, \quad s = \frac{\beta}{1 + \beta}e_1 - \frac{1}{1 + \beta} \frac{e_2}{R}.$$

Intuition: old-age income reduces required saving. Common mistake: omit e_2/R from lifetime wealth.

M2. Market clearing is $K_{t+1} = N_t s_t$. Since $k_{t+1} = K_{t+1}/N_{t+1}$ and $N_{t+1} = (1 + n)N_t$,

$$k_{t+1} = \frac{s_t}{1 + n}.$$

Common mistake: divide by N_t instead of N_{t+1} .

M3. With $k = K/(AL)$,

$$k_{t+1} = \frac{K_{t+1}}{A_{t+1}L_{t+1}} = \frac{L_t s_t}{(1 + g)(1 + n)A_t L_t} = \frac{s_t/A_t}{(1 + n)(1 + g)}.$$

Common mistake: include population growth but forget efficiency growth.

M4. Let $k = K/(AL)$. Output is $Y = ALk^\alpha$. The wage per efficiency unit is

$$W = (1 - \alpha)k^\alpha,$$

and the wage per worker is $w = AW = A(1 - \alpha)k^\alpha$. The gross return is

$$R = 1 - \delta + \alpha k^{\alpha-1}.$$

Common mistake: use the worker wage where the household budget is written in efficiency units.

M5. Government budget:

$$N_t^y \tau w_t = N_{t-1}^y p_t.$$

Since $N_t^y = (1 + n)N_{t-1}^y$, $p_t = (1 + n)\tau w_t$. Common mistake: write $p_t = \tau w_t/(1 + n)$.

14.2 Closed-form and proof drills

C1. Use $s_t = \beta w_t / (1 + \beta)$, $w_t = A_t(1 - \alpha)k_t^\alpha$, and $K_{t+1} = L_t s_t$. Divide by $A_{t+1}L_{t+1}$:

$$k_{t+1} = \frac{\beta(1 - \alpha)}{(1 + \beta)(1 + n)(1 + g)} k_t^\alpha.$$

Set $k_{t+1} = k_t = k^*$ for the formula in the catalog.

C2. Golden-rule condition is $R(k_{GR}) = (1 + n)(1 + g)$. With full depreciation and CD, $R^* = \alpha(k^*)^{\alpha-1}$. Substitute the steady state to obtain

$$R^* = \frac{\alpha(1 + \beta)(1 + n)(1 + g)}{\beta(1 - \alpha)}.$$

Dynamic inefficiency occurs when this is less than $(1 + n)(1 + g)$.

C3. Euler gives $(s/(W - s))^\sigma = \beta R^{1-\sigma}$. Therefore

$$\mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}}.$$

It rises with R if $\sigma < 1$, is flat if $\sigma = 1$, and falls if $\sigma > 1$.

C4. Compare indirect utilities. Working together multiplies labor income by $1 + h$ relative to one worker. Since two people consume in both periods, the log gain is $2(1 + \beta) \log(1 + h)$. The threshold is

$$q^*(h) = 2(1 + \beta) \log(1 + h).$$

C5. Use the two steady-state equations

$$k = a(1 - \tau)z^\lambda k^\alpha, \quad \delta_z z = \tau(1 - \alpha)z^\lambda k^\alpha.$$

Derive $z/k = \tau(1 + \beta)/[\delta_z \beta(1 - \tau)]$ and substitute into the capital equation. The tax term is proportional to $\tau^\lambda(1 - \tau)^{1-\lambda}$, so the effect changes sign around $\tau = \lambda$.

14.3 Past-quail applications

P1. Young assets are $s_t^y = -w_{t+1}/(R_{t+1}D)$; middle-aged assets are $s_t^m = \beta^2 w_t/D$. Aggregate capital is $N_t^y s_t^y + N_t^m s_t^m$. The minus sign is borrowing by young agents against middle-age income.

P2. Use the PAYG law in Model 5. The coefficient multiplying k_t^α is

$$\frac{\beta(1 - \tau)(1 - \alpha)}{(1 + n)[(1 + \beta) + \tau(1 - \alpha)/\alpha]}.$$

It falls with τ , so the unique positive steady state falls with τ .

P3. Low- γ households spend more time in the market and save more in levels. Economy A with a larger low- γ share has higher market labor and higher GDP per household. But normalized $k = K/M$ follows the same law, so the steady-state return need not be lower; in the benchmark it is the same.

P4. With type shares π_i , $e_{1i} = z_i$, and $e_{2i} = z_i(1 - \delta_i)$, loan-market clearing gives

$$0 = \sum_i \pi_i \left[\frac{\beta}{1 + \beta} z_i - \frac{1}{1 + \beta} \frac{z_i(1 - \delta_i)}{R} \right].$$

Thus

$$R = \frac{\sum_i \pi_i z_i (1 - \delta_i)}{\beta \sum_i \pi_i z_i}.$$

P5. The intratemporal FOC is $W/c_1 = A^{1-\sigma} h^{-\sigma}$. A permanent rise in A changes the right side at the old h , so h must generally adjust. The new BGP has constant hours after adjustment; it does not preserve old hours.

14.4 Trap and Economist-claim questions

T1. Wrong. Dynamic inefficiency is a return-growth comparison. College types saving less can lower capital and raise the return, which moves away from dynamic inefficiency. The test is $R^* < (1 + n)(1 + g)$.

T2. Not necessarily. More participation raises effective labor and may raise aggregate saving. If capital and effective labor scale together, normalized k^* and the rental rate can be unchanged.

T3. Wrong as stated. Nontradability changes the feasible set. Inefficiency requires a missing market or wedge relative to the relevant feasible allocation. A household producing and consuming its own non-tradable good may be constrained, but that does not by itself prove Pareto inefficiency relative to an infeasible trading allocation.

T4. Incomplete. PAYG's implicit return is the growth rate of the tax base, approximately $(1 + n)(1 + g)$. It is attractive relative to private saving only if this return compares favorably with the market return or if the economy is dynamically inefficient. Positive n alone is not enough.

T5. Physical capital chosen in the past cannot jump. A permanent TFP increase can make wages, returns, and normalized variables jump, but K_t is predetermined. Future capital K_{t+1} changes through current saving.

15 Past-Qual Practice Map

Year	Problem	Practice target	Packet sections
2016	Q2	Skilled/unskilled OLG, effective labor, skill-premium claim.	Models 2, 7.
2017	Q3	Infrastructure tax, two-state dynamics, nonmonotone tax comparative static.	Models 2, 10.
2018	Q2	Three-period setup, borrowing/saving by age, Hicks-neutral BGP.	Model 6.

Year	Problem	Practice target	Packet sections
2019	Q2	Labor profiles, old-age income, borrowing, dynamic-efficiency claim.	Models 1, 4, 7.
2020	Q3	PAYG pension formula, Social Security crowd-out, $k^*(\tau)$.	Model 5.
2021	Q1	Participation threshold, household labor supply, total versus normalized effects.	Model 8.
2022	Q2	Home production, type composition, market hours, return claim.	Model 9.
2023	Q1	Endowment OLG loan-market clearing, no-trade cases, interest comparisons.	Model 1.
2024	Q2	Non-tradable home good, BGP hours, home-productivity transition.	Model 9, Section 9.

16 Final Study Checklist

Before attempting old OLG quals, make sure you can do the following without notes.

1. Write a lifetime budget with old-age income and derive the log savings rule.
2. Aggregate individual saving into K_{t+1} and divide by the correct next-period denominator.
3. Derive Cobb-Douglas wages and returns under the problem's normalization.
4. Solve the log two-period production OLG steady state.
5. State the golden-rule condition and dynamic-efficiency test.
6. Derive PAYG pensions from the date- t government budget.
7. Handle old-age labor income as a negative term in saving.
8. Distinguish aggregate, per-worker, per-effective-worker, and per-market-worker variables.
9. Explain why young agents borrow in the three-period OLG model.
10. Solve a participation threshold using indirect utility.
11. Solve home-production time allocation and explain its effect on market labor.
12. Separate impact effects, transition effects, and new steady-state/BGP effects.
13. Answer Economist X claims with a formula first and intuition second.

17 Quality Audit

Quality gate	Status
Explicit source inventory included	Complete
Pattern analysis included before notes	Complete
Detailed blueprint included	Complete
Canonical OLG models derived	Complete
Required derivations from the manifest covered	Complete
Closed forms derived or sketched	Complete
Current problem-set style reflected where relevant	Complete
Past-qual mapping included	Complete
Common traps explicit	Complete
Exercise bank and answer sketches included	Complete
TeX uses standard packages and readable display math	Complete

Remaining caveat. The GitHub folder listed several PDFs whose raw PDF text was not extracted in this run. The packet therefore relies most heavily on accessible raw TeX lecture notes, local past-qual TeX files, the manifest, the skill file, and the uploaded local problem-set PDFs for adjacent current-course style.